

Urban Railway Devices & the Control Centre

A simple guide for non-experts: what sits beside the track, what sits along the route, and how everything talks to the Operational Control Centre (OCC)

Think of it like a city traffic control room

The train line has many “eyes and ears” next to the tracks. They send information to a central room called the **Operational Control Centre (OCC)**. People there watch screens and keep trains safe, on time, and well powered — similar to how a traffic control room watches roads using cameras and sensors.

1. Two places where equipment lives

In urban railways (metro / subway / light rail), equipment is usually grouped into two locations. People often use the words in slightly different ways, so here is the everyday meaning:

Trackside	Wayside
Equipment placed right next to or on the track — almost where the train physically passes. Examples: track sensors, signals, point machines (track switches).	Equipment placed along the railway corridor — usually in cabinets, rooms, or structures beside the line (not always touching the rail). Examples: interlocking cabinets, power rooms, radio antennas, CCTV.

*Simple memory tip: **Trackside** = next to the rails. **Wayside** = along the way / corridor (cabinets and rooms that support the line).*

2. Trackside devices (next to the rails)

These devices “see” the train, guide it, and keep the route safe. Below is what they do in plain language.

Device	What it does	Everyday analogy	How it reaches OCC
Track circuit / Axle counter	Detects whether a train is on a section of track	Like a sensor that says “this stretch of road is occupied”	Sends occupied / free status to signalling computers, then to OCC screens
Point machine (switch)	Moves the rails so a train can change tracks	Like a railway version of a road junction gate	OCC / interlocking sends “move left/right”; machine reports locked position
Signal lights	Tell the train (or driver) stop / go / caution	Like traffic lights for trains	Command comes from interlocking; aspect (colour) is reported back to OCC
Balise / Beacon / Transponder	Gives position / speed limit info to the train as it passes	Like a roadside milestone that talks to the vehicle	Works with onboard train computer; related status can be monitored via signalling to OCC
Train detection sensors / loops	Confirm train presence at platforms, crossings, or special points	Like a parking-bay sensor at a station doorway	Local controller → station system → OCC
Platform Screen Door (PSD) interface	Opens/closes glass doors on platforms in sync with the train	Like elevator doors that only open when the cabin is aligned	Door status and alarms go to station systems and OCC

Device	What it does	Everyday analogy	How it reaches OCC
Intrusion / obstacle detector	Detects people or objects on the track	Like a security alarm for the rail area	Alarm is raised immediately to OCC for action

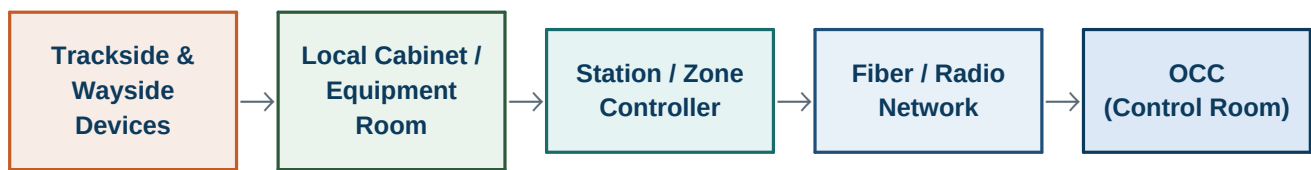
3. Wayside devices (along the corridor)

These devices support communication, power, safety, and passenger systems. They are usually in cabinets, equipment rooms, stations, or roadside structures.

Device	What it does	Everyday analogy	How it reaches OCC
Interlocking / signalling cabinet	Makes sure signals and points never create an unsafe route	Like a rules computer that prevents two trains entering the same path	Connected by fibre to zone controllers and OCC ATS / signalling desks
Zone / Station controller	Collects data from many local devices in one area	Like a neighbourhood hub that talks to city HQ	Main hop before information reaches OCC
Radio base station (TETRA / LTE-R)	Wireless voice and data between OCC, staff, and trains	Like a dedicated mobile-phone tower only for railway use	Radio network links trains/staff to OCC operators
CCTV cameras	Live video of platforms, tunnels, tracks, and stations	Like security cameras in a shopping mall	Video feeds go to OCC video wall / security desks
SCADA Remote Terminal Unit (RTU)	Reports power, ventilation, pumps, and facility status	Like a building-management box for the railway	Sends alarms and measurements to OCC SCADA screens
Traction power / substation interface	Feeds electricity to the rails (third rail / overhead)	Like a local electricity substation for the trains	Breakers, voltage, and faults monitored and controlled from OCC
Passenger Information System (PIS)	Shows next-train times and announcements	Like airport departure boards	OCC / timetable systems push messages to station displays
Emergency / Help point & PAS	Lets passengers call for help; public announcements	Like an emergency call box and PA speakers	Calls and PA control are handled by OCC / station staff
Fibre optic transmission node	High-speed data backbone along the line	Like the railway's private internet cable	Carries almost all device data into OCC

4. How everything connects to the OCC

Almost nothing talks to the OCC completely alone. Information usually travels in steps — from the device, to a nearby cabinet, to a station/zone computer, then through the railway’s private network into the OCC.



What the OCC actually receives

Information type	In plain words	Typical OCC screen / desk
Train positions & movements	Where each train is, and whether it is late or early	Automatic Train Supervision (ATS) / signalling displays
Signal & point status	Which routes are set, and whether equipment is healthy	Signalling / interlocking monitors
Power & facilities alarms	Electricity, ventilation, flooding, fire, doors	SCADA / BMS desks
CCTV & security events	Live pictures and intrusion alerts	Security / video wall
Radio & emergency calls	Talk to drivers, staff, and passengers	Radio console / help-point desk
Passenger messages	Delay notices, platform changes, announcements	PIS / PA control

5. Common connection methods (plain English)

Connection method	What it is	Typical use
Fibre optic cable	Main private data highway along tunnels and viaducts	Most signalling, SCADA, CCTV, and PIS traffic
Copper / hardwired I/O	Short local wires from a sensor to the nearby cabinet	Track circuits, door contacts, local alarms
Railway radio (TETRA / LTE-R)	Wireless link when a cable cannot reach a moving train	Driver voice, train location updates, emergency calls
Industrial Ethernet / IP network	Modern digital language used inside railway networks	Controllers, cameras, passenger displays
Serial / fieldbus links	Older but still common “device language” inside rooms	Some RTUs, legacy signalling peripherals

6. One example journey (so it clicks)

Example: a train enters a tunnel section

- 1) A **trackside axle counter** notices the train.
- 2) The nearby **wayside signalling cabinet** updates: “Section occupied.”
- 3) The **zone controller** shares that status on the fibre network.
- 4) On the **OCC wall screen**, the train icon moves into that tunnel block.
- 5) If something is wrong (for example a door alarm), an operator can radio the driver and send a passenger announcement — all from the same centre.

7. Quick glossary

OCC — Operational Control Centre — the central room that watches and manages the whole line.

ATS — Automatic Train Supervision — software that shows train movements and helps regulate service.

Interlocking — Safety logic that prevents conflicting routes (two trains cannot be given the same path).

SCADA — System that monitors and controls power and building/facility equipment.

CBTC — Communications-Based Train Control — modern metro signalling using continuous radio/data links.

Wayside / Trackside — Equipment beside the line; trackside is specifically next to the rails.

This guide is an educational overview for general understanding. Exact equipment names and architectures vary by metro project, supplier, and country standards.